

Sanoja Swain

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EDUCATION

Thapar Institute of Engineering and Technology

Bachelor of Engineering in Computer Engineering; Cumulative GPA: **8.94**(Present)

August 2023- June 2027

Patiala, Punjab

Scottish High International School

Council for the Indian School Certification Examinations; Class X percentage: 99; Class XII Percentage: 92.8

April 2023

Gurgaon, Haryana

WORK EXPERIENCE

Suvidha Foundation

13th February 2025- 13th March 2025

Machine Learning Intern

- Collaborated on a Python-based two-point text summarization system, combining **BERT** for extractive summarization and **BART** for abstractive methods.
- Curated and processed **1000+ news articles** to enhance large-scale NLP training datasets.
- Conducted literature review on PEGASUS tuning and advanced transformer architectures.

[Certification Link]: <https://tinyurl.com/SuvidhaSanojaSwain>

PROJECT WORK

AI Restaurant Reservation Agent - <https://github.com/Sanoja1606/Smart-Restaurant-Booking-Agent>

Python, OpenAI API, FastAPI, Streamlit, Pydantic, Agentic AI

- Built an **AI-powered restaurant reservation assistant** using OpenAI function calling to autonomously perform restaurant search, availability checking, and table booking through natural language conversations.
- Engineered a **3-tier architecture** (Streamlit, OpenAI Agent, FastAPI) with **2 REST APIs** and **7 backend service functions** for reservation validation, capacity management, and booking orchestration.
- Developed a conversational booking system supporting **30 restaurant listings**, enabling end-to-end reservation workflows with real-time tool execution and response generation.

AI Medical Report Analyzer- <https://health-guard-ai-li1j.vercel.app/>

FastAPI, PyTesseract, LangChain, RAG, MongoDB

- Built a **FastAPI** platform for secure medical report analysis using **JWT authentication** and **Cloudinary**.
- Developed an **RAG pipeline (LangChain + Gemini API + FAISS)** to extract parameters and provide AI-driven severity analysis.
- Engineered automated PDF generation and visual health dashboards for real-time tracking and secure family sharing.

UrbanMetrics (Streamlit Web App for House Rent Prediction)- <https://github.com/Sanoja1606/UrbanMetrics>

XGBRegressor, LGBMRegressor, Pandas, Scikit-learn, Matplotlib, Seaborn

- Developed a Streamlit web app to predict house rents based on features like location, furnishing status, number of bedrooms, and more.
- Achieved **98% accuracy** and **RMSE of 2547** using **LGBMRegressor**, delivering reliable rental price predictions.
- Conducted **Exploratory Data Analysis (EDA)** and data preprocessing to enhance model performance.
- Deployed the model using Streamlit for real-time predictions, enabling users to access rental price insights interactively.

TECHNICAL SKILLS

- Programming Languages:** C++, Python, SQL, HTML, CSS, JavaScript
- Development Tools:** Git, Visual Studio Code, Oracle Live SQL, MS Excel, Power BI
- Framework/Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, PyTorch, TensorFlow/Keras, LangChain, RAG, FastAPI, MongoDB, MySQL, SQLAlchemy
- Relevant Coursework:** Machine Learning, Data analysis, Object Oriented Programming, Data Structure and Algorithms, Database Management System

EXTRACURRICULAR ACTIVITIES

THAPAR VENTURE CLUB

Directed creative branding, promotional outreach, and comprehensive event management initiatives.

ISTE (Indian Society for Technical Education)

Hosted frontend development workshops and led the design department.

Soft Skills: Creative Thinking, Strategic planning, Adaptability, Team work | **Design Tools:** Figma, Adobe Photoshop, Adobe Illustrator